

Connectivity in the Wild

Wireless Dynamic Routing for Field IoT

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Why Field Connectivity Fails

The Field IoT Reality

Your Raspberry Pi works in the lab. The field is a different beast.



Forest fire monitoring

No cell towers, dense canopy, interference



Coastal/ocean buoys

Long distances, salty air, dynamic topology



Construction sites

Equipment moves daily, temporary networks



Precision agriculture

Km-scale fields, low power budget, remote



Disaster response

Destroyed infra, ad-hoc deployment, urgency

KEY CHALLENGES

- No fixed infrastructure
- Nodes join/leave dynamically
- Long distances + obstructions
- Tight power budgets (solar/battery)
- Heterogeneous hardware
- Intermittent connectivity
- Security in untrusted areas

Wireless Protocols for Field IoT

The Protocol Landscape — Picking the Right Radio

Range vs. power vs. bandwidth: Choose Two.

WiFi 802.11

Range ~100m
Bandwidth High
Power High
Mesh Yes (802.11s)
↳ Indoor dense deploy

Bluetooth BLE 5.x

Range ~400m
Bandwidth Low-Med
Power Very Low
Mesh Yes (BT Mesh)
↳ Wearables, short range

LoRa / LoRaWAN

Range 2–15 km
Bandwidth Very Low
Power Ultra Low
Mesh Partial (LoRa)
↳ Agriculture, tracking

Zigbee 802.15.4

Range ~100m
Bandwidth Low
Power Low
Mesh Native
↳ Building automation

Thread /Matter

Range ~100m
Bandwidth Low
Power Low
Mesh Native IPv6
↳ Smart home, edge

IEEE 802.11ah (HaLow)

Range ~1 km
Bandwidth Med
Power Low
Mesh Emerging
↳ Field sensors, IoT



Dynamic Routing in Wireless Mesh Networks

What Is Dynamic Routing?

Nodes discover each other, compute routes, and adapt — without a central controller.

Static vs. Dynamic

	Static Routing	Dynamic Routing
Configuration	Manual, fixed	Auto-discovered
Topology change	Breaks network	Self-heals
Overhead	None	Protocol control traffic
Scalability	Poor	Good
Field suitability	✗	✓

Key Routing Paradigms

Proactive (table-driven)

Each node maintains full topology table. E.g., OLSR, DSDV. Low latency, higher overhead.

Reactive (on-demand)

Routes discovered only when needed. E.g., AODV, DSR. Lower overhead, higher latency.

Hybrid

Proactive in local zones, reactive across. E.g., ZRP, BATMAN. Balances both.

Routing Protocols You'll Encounter

Each designed with different trade-offs for field environments.

OLSR

Optimized Link-State Routing

Type: Proactive | RFC 3626 / 7181

- Uses MPR (Multipoint Relays) to reduce flooding
- Maintains full topology — fast route lookup
- Good for dense, stable meshes
- Used in: community Wi-Fi, military radios

AODV

Ad-hoc On-demand Distance Vector

Type: Reactive | RFC 3561

- RREQ/RREP flood to discover routes
- Routes expire when unused — low overhead
- Suitable for sparse, low-traffic field nets
- Used in: sensor networks, disaster comms

B.A.T.M.A.N.

Better Approach To Mobile Adhoc Networking

Type: Hybrid | RFC 7528

- Distributes routing knowledge — no single node knows full topology
- OGM (Originator Messages) flood to rank neighbors
- batman-adv: kernel-level Linux implementation
- Used in: Freifunk, community mesh networks



Progress & Emerging Research

Where the Field Is Heading

Recent breakthroughs making field IoT more reliable and scalable.

75B+

IoT devices by 2025

~\$1B

DARPA mesh research

802.11s

IEEE mesh std (2012+)

HaLow

Wi-Fi IoT at 900MHz

ML-Assisted Routing

Neural networks predict link quality and pre-adapt routes before failures occur. Projects: DeepRoute, RL-OLSR. Reduces packet loss in mobile meshes by 30–60%.

Energy-Aware Routing

Route selection weighted by remaining battery, solar harvest rate. Research: EAODV, Green Mesh. Critical for multi-year deployments without maintenance.

Delay-Tolerant Networking (DTN)

Store-carry-forward for disconnected environments. NASA uses DTN for deep-space comms. Terrestrial use: wildlife tracking, remote sensors. Bundle Protocol (RFC 5050/9171).

SDN for Mesh IoT

Software-Defined Networking decouples control from data plane in mesh networks. Centralized view for optimization; distributes on controller loss. Research: SD-MANET.

Real-World Deployments

Field IoT mesh networks already in production at scale.



Sonoma County Fire Mesh

Sonoma County, CA + UC Berkeley BETS Lab

Tech: LoRa + 802.11s Wi-Fi mesh, solar-powered

Early wildfire detection across 1,600 sq miles. Nodes relay data hop-by-hop to satellite uplinks when no cell coverage.



Freifunk Community Networks

Germany + 400+ cities globally

Tech: 802.11s / OLSR / batman-adv on commodity hardware

Open community mesh networks providing free internet access. 50,000+ nodes across Europe. Survived ISP outages.



PrecisionHawk Agri-IoT

Agriculture sector, deployed globally

Tech: LoRaWAN + cellular fallback, multi-hop relay

Soil moisture, crop health sensors networked across km-scale farms. Nodes communicate via AODV-style relays.



DARPA OFFensive Swarm-Enabled Tactics (OFFSET)

DARPA / U.S. DoD

Tech: Dynamic mesh, adaptive routing, heterogeneous radios

250-drone swarms that self-organize mesh communication networks. Routing adapts in ms as units move.



Hands-On Code & Frameworks

Get Your Hands Dirty – Frameworks & Tools

All open-source; several run directly on Raspberry Pi & ESP32 (your course hardware!).

batman-adv

Kernel-level mesh routing for Linux

HW: Raspberry Pi 

```
sudo apt install batctl
modprobe batman-adv
batctl if add wlan0
# Full mesh in ~5 min!
```

Runs on RPi's kernel. Connect multiple RPis via Wi-Fi and create a real mesh.



www.open-mesh.org

Meshtastic

LoRa mesh firmware + Python API

HW: ESP32  + RPi 

```
pip install meshtastic
# Flash ESP32 via meshtastic-flasher
meshtastic --sendtext 'Hello mesh!'
# Works on ESP32 + Python client on RPi
```

Flash your ESP32 in class with LoRa mesh. Python API for data collection on RPi.



meshtastic.org

OpenWRT + OLSR

Router OS + proactive routing daemon

HW: Linux / RPi

```
# On OpenWRT router or RPi:
sudo apt install olsrd
olsrd -f /etc/olsrd/olsrd.conf -d 2
# Watch topology in real-time
```

Industry-standard proactive routing. Visualize the mesh graph as it forms.



olsr.org / openwrt.org

Simulation, Learning & Going Deeper

Before you deploy hardware, simulate — then learn from the best.

Simulation Tools

ns-3

Gold standard network simulator. Has WiFi mesh, LoRa, OLSR, AODV modules. Python scripting.

Cooja (Contiki)

IoT-specific simulator for low-power mesh nodes. Simulates 100s of motes running RPL routing.

GNS3 + OpenWRT

Emulate real router OSES with OLSR/BATMAN. Visual topology editor. Great for protocol experiments.

Papers & Learning Resources

Paper

Perkins & Royer (1999): AODV — the original foundational reactive routing paper. IEEE WMCSA.

Paper

Johnson & Maltz (1996): DSR dynamic source routing in ad-hoc wireless networks. Mobicom.

Course

Coursera: 'Internet of Things: Sensing & Actuation' (UC San Diego) — mesh networking module.

Repo

<https://github.com/meshtastic/firmware> — real production LoRa mesh firmware for ESP32.

<https://git.open-mesh.org/batman-adv.git>, <https://git.open-mesh.org/batctl.git>, https://www.open-mesh.org/projects/batman-adv/wiki/BATMAN_IV — Linux kernel mesh module. Read the code, run the tests.

Try It: Build a 3-Node Mesh on Your Course Hardware

A weekend project using your Raspberry Pi + MQTT skills from this course.

01 Flash & Setup

Install `batman-adv` on 3 Raspberry Pis. Configure `wlan0` as ad-hoc interface. No router needed — direct peer-to-peer.

02 Form the Mesh

Run `batctl` to verify neighbors. Use ``batctl o`` to see originator table. Watch nodes find each other automatically.

03 MQTT Over Mesh

Run Mosquitto broker on one node. Publish sensor data from the others. Watch messages hop through intermediate nodes.

04 Break & Heal

Kill one Pi mid-transfer. Watch BATMAN re-route through surviving nodes. Log failover time. This is dynamic routing in action!

Key Takeaways

- 1 Field IoT $\neq$ lab IoT — topology changes, infrastructure disappears, power is scarce
- 2 Protocol choice matters: LoRa for km-scale low-power, WiFi mesh for dense short-range, LoRa+mesh (Meshtastic) for the middle ground
- 3 Dynamic routing (BATMAN, OLSR, AODV) enables self-healing networks without admin intervention
- 4 ML, DTN, and energy-aware routing are the frontier — fertile ground for research and projects
- 5 You already have the hardware (RPI + ESP32) — a 3-node mesh is a weekend project away!

Questions?

Quick Links:

↳ meshtastic.org

↳ open-mesh.org/batman-adv

↳ olsr.org

↳ ns-3.org

↳ github.com/meshtastic

↳ github.com/open-mesh

Thank you — and happy meshing! 📶